

Matrix arithmetic

- **Exercise 1**

Let

$$A = \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ 3 & -2 \end{bmatrix}.$$

Compute $A + B$, $2A - 3B$, AB , BA .

- **Solution 1.1**

We have that

$$\begin{aligned} A + B &= \begin{bmatrix} 6 & -1 \\ 3 & 2 \end{bmatrix}, \\ 2A - 3B &= \begin{bmatrix} -13 & -7 \\ -9 & 14 \end{bmatrix}, \\ AB &= \begin{bmatrix} -1 & 5 \\ 12 & -8 \end{bmatrix}, \\ BA &= \begin{bmatrix} 5 & -6 \\ 3 & -14 \end{bmatrix}. \end{aligned}$$

- **Exercise 2**

Find the size of the product matrix.

- **Solution 2.1**

$$\begin{aligned} (2 \times 3)(3 \times 4) &\implies (2 \times 4). \\ (4 \times 1)(1 \times 2) &\implies (4 \times 2). \\ (1 \times 2)(3 \times 1) &\implies \text{Not defined.} \\ (2 \times 2)(2 \times 4) &\implies (2 \times 4). \end{aligned}$$

- **Exercise 3**

Suppose $A + B$ and AB are defined. What can you say about the sizes of A and B ?

- **Solution 3.1**

As $A + B$ is well-defined, both matrices have the same size $n \times m$.

As AB is well-defined, $m = n$. Therefore A, B have size $n \times n$.

- **Exercise 4 (Recovering a matrix from its action)**

Suppose $A \in M_{2 \times 3}(\mathbb{R})$ satisfies

$$A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \quad A \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.$$

1. Determine A .

2. Compute

$$A \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix}$$

3. In general, what information about a matrix A is contained in $Ae_1, Ae_2, \dots, Ae_m$, $A \in M_{m \times n}(\mathbb{R})$?

- **Solution 4.1**

1. Let $A = [a_{ij}]_{2 \times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$. By the definition, we have

$$\begin{aligned}
A \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} &= \begin{bmatrix} a_{11} \cdot 1 + a_{12} \cdot 0 + a_{13} \cdot 0 \\ a_{21} \cdot 1 + a_{22} \cdot 0 + a_{23} \cdot 0 \end{bmatrix} = \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \\
A \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} &= \begin{bmatrix} a_{11} \cdot 0 + a_{12} \cdot 1 + a_{13} \cdot 0 \\ a_{21} \cdot 0 + a_{22} \cdot 1 + a_{23} \cdot 0 \end{bmatrix} = \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \\
A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} &= \begin{bmatrix} a_{11} \cdot 0 + a_{12} \cdot 0 + a_{13} \cdot 1 \\ a_{21} \cdot 0 + a_{22} \cdot 0 + a_{23} \cdot 1 \end{bmatrix} = \begin{bmatrix} a_{13} \\ a_{23} \end{bmatrix} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}.
\end{aligned}$$

Therefore, we determine the matrix A to be

$$A = \begin{bmatrix} 2 & 3 & -1 \\ -1 & 2 & 4 \end{bmatrix}.$$

2. By using the recovered matrix A , we directly compute the product

$$A \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix} = \begin{bmatrix} 2 & 3 & -1 \\ -1 & 2 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ -5 \end{bmatrix} = \begin{bmatrix} 2 \cdot 2 + 3 \cdot 3 + (-1) \cdot (-5) \\ (-1) \cdot 2 + 2 \cdot 3 + 4 \cdot (-5) \end{bmatrix} = \begin{bmatrix} 4 + 9 + 5 \\ -2 + 6 - 20 \end{bmatrix} = \begin{bmatrix} 18 \\ -16 \end{bmatrix}.$$

3. Let $A = [a_{ij}]_{m \times n}$ and let $e_k = [\delta_{jk}]_{n \times 1}$ be the k -th standard basis vector where $\delta_{jk} = 1$ if $j = k$ and $\delta_{jk} = 0$ if $j \neq k$. Then the i -th entry of the column vector Ae_k is given by

$$\sum_{j=1}^n a_{ij} \delta_{jk} = a_{ik}.$$

Thus, we obtain the algebraic identity

$$Ae_k = \begin{bmatrix} a_{1k} \\ a_{2k} \\ \vdots \\ a_{mk} \end{bmatrix} = C_k(A).$$

Therefore, for each $k \in \{1, 2, \dots, n\}$, the product Ae_k contains exactly the k -th column of the matrix A .

• Exercise 5

Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. Show that if $A \in M_{m \times n}(\mathbb{F})$ satisfies $Ax = 0$ for all $x \in \mathbb{F}^n$, then $A = 0$. Deduce that if $Ax = Bx$ for all $x \in \mathbb{F}^n$, then $A = B$.

◦ Proof 5.1

Suppose $A \in M_{m \times n}(\mathbb{F})$ with the property that $\forall x \in \mathbb{F}^n, Ax = 0$. Let $e_j = [\delta_{kj}]_{n \times 1}$ be the j -th standard basis vector of $\mathbb{F}^n$ for each $1 \leq j \leq n$. By hypothesis, we have

$$Ae_j = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}_{m \times 1}.$$

By Exercise 4.3, the product Ae_j represents the j -th column of A , which yields $C_j(A) = 0$ for all $1 \leq j \leq n$. Since all columns of A are zero, we obtain $A = 0$.

Now, suppose $\forall x \in \mathbb{F}^n, Ax = Bx \iff Ax - Bx = 0$. By using Theorem 7.3, this can be rewritten as

$$(A - B)x = 0 \implies A - B = 0 \implies A = B.$$

• Exercise 6

- Find nonzero matrices $A \in M_{2 \times 3}(\mathbb{R})$ and $B \in M_{3 \times 2}(\mathbb{R})$ such that $AB = 0_{2 \times 2}$ but $BA \neq 0_{3 \times 3}$.
- Use your example to show that the following statements are false:

- $AB = 0 \implies A = 0 \vee B = 0.$
- $AB = 0 \implies BA = 0.$

◦ **Solution 6.1**

1. Let the non-zero matrices be defined as

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{2 \times 3}, \quad B = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 0 \end{bmatrix}_{3 \times 2}.$$

By the definition of matrix multiplication, we compute the product AB as

$$AB = \begin{bmatrix} 1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot (-1) + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \mathbb{0}_{2 \times 2}.$$

Now, we compute the product BA as

$$BA = \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 0 \cdot 0 \\ (-1) \cdot 1 + 0 \cdot 0 & (-1) \cdot 1 + 0 \cdot 0 & (-1) \cdot 0 + 0 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ -1 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \neq \mathbb{0}_{3 \times 3}.$$

2. By using the explicit counterexample constructed in part (1), we analyze the truth values of the statements:

- Since $A \neq \mathbb{0}_{2 \times 3}$ and $B \neq \mathbb{0}_{3 \times 2}$ but $AB = \mathbb{0}_{2 \times 2}$, the implication $AB = 0 \implies A = 0 \vee B = 0$ is false.
- Since $AB = \mathbb{0}_{2 \times 2}$ but $BA \neq \mathbb{0}_{3 \times 3}$, the implication $AB = 0 \implies BA = 0$ is false.

• **Exercise 6 (Which familiar scalar identities survive?)**

Let $A, B \in M_n(\mathbb{R})$.

1. Expand $(A + B)^2$ using only distributivity and associativity.
2. Show that

$$(A + B)^2 = A^2 + 2AB + B^2$$

holds if and only if $AB = BA$.

3. Prove that

$$(A - B)(A + B) = A^2 - B^2$$

holds if and only if $AB = BA$.

4. Give explicit 2×2 matrices for which both familiar scalar formulas fail.

◦ **Solution 6.1**

1. By applying the distributive law and associativity of matrix multiplication, we expand the expression as

$$\begin{aligned} (A + B)^2 &= (A + B)(A + B) \\ &= (A + B)A + (A + B)B \\ &= AA + BA + AB + BB \\ &= A^2 + BA + AB + B^2. \end{aligned}$$

2. We establish the equivalence by comparing the algebraic expansions

$$\begin{aligned} (A + B)^2 = A^2 + 2AB + B^2 &\iff A^2 + BA + AB + B^2 = A^2 + 2AB + B^2 \\ &\iff BA + AB = 2AB \\ &\iff BA + AB = AB + AB \\ &\iff BA = AB. \end{aligned}$$

3. By expanding the left side using the distributive law, we have

$$\begin{aligned}
(A - B)(A + B) &= (A - B)A + (A - B)B \\
&= AA - BA + AB - BB \\
&= A^2 - BA + AB - B^2.
\end{aligned}$$

We establish the equivalence by comparing the algebraic expansions

$$\begin{aligned}
(A - B)(A + B) = A^2 - B^2 &\Leftrightarrow A^2 - BA + AB - B^2 = A^2 - B^2 \\
&\Leftrightarrow -BA + AB = 0 \\
&\Leftrightarrow AB = BA.
\end{aligned}$$

4. Let the two non-commuting 2×2 matrices be defined as

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

We verify their products directly by the definition of matrix multiplication

$$\begin{aligned}
AB &= \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 \cdot 0 + 0 \cdot 0 & 1 \cdot 1 + 0 \cdot 0 \\ 0 \cdot 0 + 0 \cdot 0 & 0 \cdot 1 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}. \\
BA &= \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 0 \\ 0 \cdot 1 + 0 \cdot 0 & 0 \cdot 0 + 0 \cdot 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.
\end{aligned}$$

Since $AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = BA$, the condition $AB = BA$ is false. By parts 2 and 3, both scalar formulas fail for these explicit matrices.